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Battery Module Including Module Case and Battery Pack Including Battery Module

Abstract

A battery module includes a cell assembly including a plurality of battery cells, and a module case accommodating the cell assembly. The module case includes an accommodation portion including a main plate supporting the cell assembly and a sidewall extending from the main plate, a module cover including an upper plate covering the cell assembly, the upper plate having a first surface oriented in a first direction, a sidewall cover extending from the upper plate, the sidewall cover covering a portion of the sidewall, the sidewall cover having a second surface oriented in a second direction opposite to the first direction, and a guide groove formed in the second surface, and a first bonding portion connecting the second surface of the sidewall cover and the sidewall to each other.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This patent document claims the priority and benefits of Korean Patent Application No. 10-2024-0029828 filed on Feb. 29, 2024, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a battery module including a module case and a battery pack including the same.

BACKGROUND

[0003] Unlike primary batteries, secondary batteries may be charged and discharged, and thus may be applied to devices within various fields, such as digital cameras, mobile phones, laptops, hybrid vehicles, electric vehicles, and energy storage systems. Secondary batteries include lithium-ion batteries, nickel-cadmium batteries, nickel-metal hydride batteries, or nickel-hydrogen batteries.

[0004] Lithium secondary batteries may be manufactured as pouch-type battery cells having flexibility, prismatic battery cells having rigidity, or cylindrical can-type battery cells. A plurality of battery cells may be stacked to form a cell assembly.

[0005] A cell assembly may be disposed in a case to form a battery module, and a plurality of battery modules may be disposed in a pack housing to form a battery pack.

SUMMARY

[0006] A battery module may include a cell assembly and a module case accommodating the cell assembly. The module case may include a plurality of components (for example, a module cover, an accommodation portion, and end plate). The components of the module case may be welded to each other and manufactured. However, when the module cover and the accommodation portion are formed of the same material, a contact surface (for example, a welded portion) between the module cover and the accommodation portion may not be easily detected. During a process of welding the module case, generated heat may be transferred to an unintended point, and the battery module may be damaged. A gap may be formed between the components of the module case, such that the module case may have reduced durability.

[0007] According to an aspect of the present disclosure, a battery module may have improved manufacturing convenience.

[0008] According to an aspect of the present disclosure, a battery module may prevent damage to a module case and a cell assembly during a welding process.

[0009] According to an aspect of the present disclosure, a battery module may have a reduced gap between components of a module case, and may have reduced durability.

[0010] A battery module and a battery pack of the present disclosure may be widely applied in the field of green technology, such as to electric vehicles, battery charging stations, and other battery-utilizing solar power generation schemes, wind power generation schemes, or the like. In addition, the battery module and the battery pack of the present disclosure may be used in eco-friendly electric vehicles, hybrid vehicles, or the like, to prevent climate change by mitigating the effects of air pollution and greenhouse gas emissions.

[0011] According to an aspect of the present disclosure, there is provided a battery module including a cell assembly including a plurality of battery cells, and a module case accommodating the cell assembly. The module case may include an accommodation portion including a main plate supporting the cell assembly and a sidewall extending from the main plate, a module cover including an upper plate covering the cell assembly, the upper plate having a first surface oriented

in a first direction, a sidewall cover extending from the upper plate, the sidewall cover covering a portion of the sidewall, the sidewall cover having a second surface oriented in a second direction opposite to the first direction, and a guide groove formed in the second surface, and a first bonding portion connecting the second surface of the sidewall cover and the sidewall to each other.

[0012] The sidewall cover may have a shade forming surface, parallel to the second surface, the shade forming surface forming at least a portion of the guide groove. The shade forming surface may be spaced apart from the first bonding portion.

[0013] The sidewall may include a guide region formed by the sidewall cover, the guide region having at least a portion exposed through the guide groove, and an exposed region spaced apart from the guide region, the exposed region not covered by the module cover. The guide region may have a first brightness, and the exposed region may have a second brightness, different from the first brightness.

[0014] The first bonding portion may include a plurality of first bonding portions. The guide region may be positioned between the plurality of first bonding portions.

[0015] A material of the module cover and a material of the accommodation portion may be the same.

[0016] The module case may include an end plate covering at least a portion of the cell assembly, the end plate perpendicular to the sidewall. The main plate may include a first central portion formed to have a first thickness, and a first end portion extending from the first central portion, the first end portion formed to have a second thickness, less than the first thickness. The end plate may include a second central portion formed to have a third thickness, and a second end portion extending from the second central portion, the second end portion formed to have a fourth thickness, less than the third thickness. The module case may include a second bonding portion connecting the first end portion and the second end portion to each other.

[0017] At least a portion of the first end portion may have a first internal surface opposing the second central portion, and a first bonding surface perpendicular to the first internal surface. The second end portion may have a second internal surface opposing the first central portion, and a second bonding surface perpendicular to the second internal surface and the first bonding surface. The second bonding portion may connect the first bonding surface and the second bonding surface to each other.

[0018] The second bonding portion may be formed in a welding direction, inclined with respect to the first bonding surface and the second bonding surface.

[0019] The upper plate may include a first end region connected to the sidewall cover, and a second end region perpendicular to the first end region. The module case may include an end plate covering at least a portion of the cell assembly, the end plate perpendicular to the sidewall, and a third bonding portion connecting the second end region of the upper plate and the end plate to each other.

[0020] The end plate may have a third surface oriented in the first direction. The third bonding portion may connect the first surface of the upper plate and the third surface of the end plate to each other.

[0021] According to another aspect of the present disclosure, there is provided a battery module including a cell assembly including a plurality of battery cells, and a module case accommodating the cell assembly. The module case may include an accommodation portion including a main plate supporting the cell assembly, and a sidewall extending from the main plate, a module cover including an upper plate covering the cell assembly, the upper plate having a first surface oriented in a first direction, and a sidewall cover extending from the upper plate, the sidewall cover covering a portion of the sidewall, the sidewall having a second surface oriented in a second direction opposite to the first direction, an end plate surrounding at least a portion of the cell assembly, the end plate having a third surface oriented in the first direction, a first bonding portion connecting the second surface of the sidewall cover and the sidewall to each other, a second bonding portion

connecting the main plate and the end plate to each other, and a third bonding portion connecting the first surface of the upper plate and the third surface of the end plate to each other.

[0022] The module cover may include a guide groove formed in the second surface. The sidewall cover may have a shade forming surface, parallel to the second surface, the shade forming surface forming at least a portion of the guide groove. The shade forming surface may be spaced apart from the first bonding portion.

[0023] The end plate may be perpendicular to the sidewall. The main plate may include a first central portion formed to have a first thickness, and a first end portion extending from the first central portion, the first end portion formed to have a second thickness, less than the first thickness. The end plate may include a second central portion formed to have a third thickness, and a second end portion extending from the second central portion, the second end portion formed to have a fourth thickness, less than the third thickness. The second bonding portion may connect the first end portion and the second end portion to each other.

[0024] At least a portion of the first end portion may have a first internal surface opposing the second central portion, and a first bonding surface perpendicular to the first internal surface. The second end portion may have a second internal surface opposing the first central portion, and a second bonding surface perpendicular to the second internal surface and the first bonding surface. The second bonding portion may connect the first bonding surface and the second bonding surface to each other.

[0025] According to another aspect of the present disclosure, there is provided a battery pack including a plurality of battery modules, and a pack frame accommodating the plurality of battery modules. Each of the plurality of battery modules may include a cell assembly including a plurality of battery cells, and a module case accommodating the cell assembly. The module case may include an accommodation portion including a main plate supporting the cell assembly, and a sidewall extending from the main plate, a module cover including an upper plate covering the cell assembly, the upper plate having a first surface oriented in a first direction, a sidewall cover extending from the upper plate, the sidewall cover covering a portion of the sidewall, the sidewall cover having a second surface oriented in a second direction opposite to the first direction, and a guide groove formed in the second surface, and a first bonding portion connecting the second surface of the sidewall cover and the sidewall to each other.

[0026] According to an embodiment of the present disclosure, manufacturing convenience of a battery module may be increased during a welding process.

[0027] According to an embodiment of the present disclosure, damage to a battery module during a welding process may be prevented.

[0028] According to an embodiment of the present disclosure, a gap between components of a module housing may be reduced, and durability of a battery module may be increased.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0029] Certain aspects, features, and advantages of the disclosed technology are illustrated by the following detailed description with reference to the accompanying

DRAWINGS

[0030] FIG. 1 is a perspective view of a battery cell according to an embodiment.

[0031] FIG. 2 is a perspective view of a battery module according to an embodiment.

[0032] FIG. 3 is an exploded perspective view of a battery module according to an embodiment.

[0033] FIG. 4 is an exploded perspective view of a module case according to an embodiment.

[0034] FIG. 5 is a side view of a battery module according to an embodiment.

[0035] FIG. 6 is an enlarged view of region “A” of FIG. 5 according to an embodiment.

[0036] FIG. 7A is a perspective view of region “B” of FIG. 6 according to an embodiment.

[0037] FIG. 7B is a cross-sectional perspective view of region “B” of FIG. 6 according to an embodiment.

[0038] FIG. 8 is a cross-sectional view taken along line A-A’ of FIG. 2 according to an embodiment.

[0039] FIG. 9 is a top view of a battery module according to an embodiment.

[0040] FIG. 10 is an exploded perspective view of a battery pack according to an embodiment.

DETAILED DESCRIPTION

[0041] Hereinafter, the present disclosure will be described in detail with reference to the accompanying drawings. However, embodiments are merely exemplary, and the present disclosure is not limited to specific embodiments.

[0042] The words and terminologies used in the specification and claims should not be construed with common or dictionary meanings, but construed as meanings and conception coinciding the spirit of the present disclosure under a principle that the inventor(s) may appropriately define the conception of the terminologies to explain the present disclosure in the optimum method.

[0043] Therefore, embodiments described in the specification and the configurations illustrated in the drawings are not more than the most preferred embodiments of the present disclosure and do not fully cover the spirit of the present disclosure. Accordingly, it should be understood that there may be various equivalents and modifications that may replace those when the present application is filed.

[0044] In addition, detailed descriptions of well-known functions and configurations that may obscure the gist of the present disclosure will be omitted. In addition, some components are exaggerated, omitted, or schematically illustrated in the accompanying drawings, and the size of each component does not fully reflect the actual size.

[0045] FIG. 1 is a perspective view of a battery cell according to an embodiment.

[0046] Referring to FIG. 1, the battery cell **100** may include a pouch **110**, an electrode assembly **120**, and an electrode tab **130**. The battery cell **100** may be a secondary battery. For example, the battery cell **100** may be a lithium-ion battery, but the present disclosure is not limited thereto. For example, the battery cell **100** may be a nickel-cadmium electricity, a nickel-metal hydride battery, or a nickel-hydrogen battery that may be charged and discharged.

[0047] The pouch **110** may form at least a portion of an exterior of the battery cell **100**. The pouch **110** may include an electrode accommodation portion **111** accommodating the electrode assembly **120**, and a sealing portion **115** sealing at least a portion of a circumference of the electrode accommodation portion **111**. The electrode accommodation portion **111** may provide a space in which the electrode assembly **120** and an electrolyte are accommodated.

[0048] The sealing portion **115** may be formed by bonding of at least a portion of a circumference of the pouch **110**. The sealing portion **115** may be in the form of a flange extending outwardly from the electrode accommodation portion **111** that is in the form of a container, and may be disposed along at least a portion of an outer periphery of the electrode accommodation portion **111**. In an embodiment, the sealing portion **115** may include a first sealing portion **115a** in which the electrode tab **130** is positioned, and a second sealing portion **115b** in which the electrode tab **130** is not positioned. A portion of the electrode tab **130** may be withdrawn or exposed to the outside of the pouch **110**. In a position in which the electrode tab **130** is drawn out, in order to simultaneously increase a sealing degree of the first sealing portion **115a** and ensure an electrical insulation state, the electrode tab **130** may be in a state of being covered by an insulating film **140**. The insulating film **140** may be formed of a film material, thinner than that of the electrode tab **130**, and may be attached to both surfaces of the electrode tab **130**.

[0049] The electrode tab **130** may transfer a current of the battery cell **100** to the outside of the battery cell **100**. The electrode tab **130** may be connected to the electrode assembly **120**.

[0050] In an embodiment, the electrode tabs **130** may be disposed at both sides of the battery cell

100 in a length direction (Y-axis direction) to be oriented in opposite directions. For example, the electrode tab **130** may include a first electrode tab **130a** (for example, a cathode) having a first polarity (for example, a cathode) toward one side of the battery cell **100** in the length direction, and a second electrode tab **130b** having a second polarity (for example, an anode) toward the other side of the battery cell **100** in the length direction. In the embodiment illustrated in FIG. 1, the sealing portion **115** may include two first sealing portions **115a** on which the electrode tab **130** is disposed and one second sealing portion **115b** on which the electrode tab **130** is not disposed. The first sealing portion **115a** may seal at least a portion of the electrode tab **130**. In an embodiment, the electrode tab **130** may be referred to as an electrode lead.

[0051] A direction in which the electrode tab **130** is positioned may be selectively designed. In an embodiment (for example, FIG. 1), the electrode tab **130** may include the first electrode tab **130a**, and the second electrode tab **130b** positioned in an opposite direction of the first electrode tab **130a** with respect to the electrode assembly **120**. In FIG. 1, it is illustrated that the electrode tabs **130** is disposed at both sides of the battery cell **100** in the length direction (for example, the Y-axis direction) to be oriented in opposite directions, but a structure of the electrode tabs **130** is not limited thereto. For example, two electrode tabs **130** may be arranged to be substantially parallel to each other in the length direction (for example, the Y-axis direction) of the battery cell **100**.

[0052] As illustrated in FIG. 1, the pouch **110** is not limited to having a structure in which the sealing portion **115** is formed on three surfaces obtained by folding one sheet of exterior material.

[0053] In an embodiment of the present disclosure, at least a portion of the sealing portion **115** may be in the form of being folded at least once. At least a portion of the sealing portion **115** may be folded, thereby improving bonding reliability of the sealing portion **115** and minimizing an area of the sealing portion **115**. The second sealing portion **115b** of the sealing portion **115** according to an embodiment, on which the electrode tab **130** is not disposed, may be folded twice and then fixed by an adhesive member (not illustrated). An angle or the number of times the second sealing portion **115b** being bent may be changed. For example, in an embodiment not illustrated, the second sealing portion **115b** may be folded by 90° with respect to the first sealing portion **115a**.

[0054] The electrode assembly **120** may include a cathode plate, an anode plate, and a separator. The separator may prevent contact between the cathode plate and the anode plate. It will be understood by those skilled in the art that the electrode assembly **120** may be manufactured using various methods. According to embodiments, a cathode, an anode, and a separator may be repeatedly disposed to form an electrode assembly. In some embodiments, the electrode assembly may be a winding-type electrode assembly, a stacking-type electrode assembly, a Z-folding-type electrode assembly, or a stack-folding-type electrode assembly.

[0055] FIG. 2 is a perspective view of a battery module according to an embodiment. FIG. 3 is an exploded perspective view of a battery module according to an embodiment. FIG. 4 is an exploded perspective view of a module case according to an embodiment.

[0056] Referring to FIGS. 2, 3, and/or 4, a battery module **200** may include a cell assembly **101**, a busbar assembly **210**, a sensor assembly **220**, an insulating plate **240**, and/or a module case **300**.

[0057] The cell assembly **101** may include a plurality of battery cells **100**. The description of the battery cell **100** of FIG. 1 may be applied to the battery cell **100** of FIG. 3.

[0058] The cell assembly **101** may have a substantially hexahedral shape. In an embodiment, the cell assembly **101** may be referred to as a cell stack. In an embodiment, the cell assembly **101** may include a plurality of battery cells **100**, connected to each other using an adhesive tape. According to an embodiment, the cell assembly **101** may include a heat propagation prevention member positioned between at least some battery cells **100**, among the plurality of battery cells **100**.

[0059] The busbar assembly **210** may include an electrically conductive internal busbar **211** electrically connected to the electrode tab **130** of the battery cell **100**, and a busbar frame **212** supporting the internal busbar **211**. The busbar frame **212** may be formed of an electrically insulating material (for example, a polymer). The busbar frame **212** may include at least one

fastening hole for accommodating a coupling component (for example, a screw, a rivet, and/or a boss structure). The busbar frame **212** may be coupled to the module case **300** using the coupling component. In an embodiment, the internal busbar **211** may be referred to as a busbar. The internal busbar **211** may be provided as a plurality of internal busbars **211**. For example, the number of internal busbars **211** may be selectively designed based on the number of battery cells **100**, included in the battery module **200**.

[0060] The busbar assembly **210** may include at least one terminal busbar **213** for electrical connection with the outside. The electrode tab **130** of the battery cell **100** may be electrically connected to the outside of the battery module **200** through the internal busbar **211** and the terminal busbar **213**. For example, the terminal busbar **213** may be electrically connected to the internal busbar **211**, and current of the battery cell **100** may be transferred to the outside of the battery module **200** through the internal busbar **211** and the terminal busbar **213**. The terminal busbar **213** may be exposed to the outside of the module case **300** through a hole **250a** of the insulating plate **240**. In an embodiment, the terminal busbar **213** may be referred to as a high-voltage busbar or a long busbar.

[0061] The sensor assembly **220** may include a sensor, sensing information on the battery module **200**. For example, the sensor assembly **220** may include a temperature sensor, sensing a temperature of the battery cell **100**, and at least one voltage sensing terminal connected to the internal busbar **211**.

[0062] The insulating plate **240** may prevent contact between at least a portion (for example, the terminal busbar **213**) of the busbar assembly **210** and the module case **300** (for example, the accommodation portion **301**). For example, at least a portion of the insulating plate **240** may be disposed between the terminal busbar **213** and a sidewall **320** of the accommodation portion **301**. The insulating plate **240** may be formed of an insulating material (for example, a polymer), and may prevent electric communication due to contact between the terminal busbar **213** and the sidewall **320**. In an embodiment, the insulating plate **240** may be disposed between the cell assembly **101** and the sidewall **320**.

[0063] In an embodiment, the battery module **200** may include a heat radiation member **250**. The heat radiation member **250** may be positioned between the cell assembly **101** and the accommodation portion **301** (for example, a main plate **310**) of the module case **300**. The heat radiation member **250** may be a thermal adhesive. At least a portion of heat generated by the battery cell **100** of the cell assembly **101** may be transferred to the accommodation portion **301** through the heat radiation member **250**.

[0064] The module case **300** may form at least a portion of an exterior of the battery module **200**. The module case **300** may accommodate a component (for example, the cell assembly **101**) of the battery module **200**. For example, the module case **300** may form an accommodation space for accommodating the cell assembly **101** and/or the busbar assembly **210**. The module case **300** may protect the cell assembly **101** from external impacts.

[0065] The module case **300** may include the main plate **310** supporting the cell assembly **101**, and the accommodation portion **301** including the sidewall **320** extending from the main plate **310**. In an embodiment, the main plate **310** may be referred to as a lower plate. The cell assembly **101** may be mounted on the main plate **310**. The sidewall **320** may cover at least a portion of a side surface of the cell assembly **101**. For example, the sidewall **320** may extend from both end portions of the main plate **310** in a first direction (for example, +Z-direction). In an embodiment, the sidewall **320** may be formed integrally with the main plate **310**.

[0066] The module case **300** may include a module cover **302** covering the cell assembly **101**. For example, the module cover **302** may be disposed at one side of the cell assembly **101**. The module cover **302** may cover a portion (for example, an upper portion) of the cell assembly **101**. The module cover **302** may protect the cell assembly **101** from external impacts of the battery module **200**. In an embodiment, the cover **230** may be referred to as an upper cover.

[0067] The module cover **302** may include an upper plate **330** covering the cell assembly **101**, and a sidewall cover **340** extending from the upper plate **330** in a second direction (for example, $-Z$ -direction). The upper plate **330** may have a first surface **330a** oriented in the first direction (for example, $+Z$ -direction). The sidewall cover **340** may extend from both end portions of the upper plate **330** in the second direction (for example, $-Z$ -direction). The sidewall cover **340** may oppose a portion of the sidewall **320** of the accommodating portion **301**, and may cover a portion of the sidewall **320**.

[0068] The module case **300** may be formed of a material that may be bonded using welding. According to an embodiment, the module case **300** may be formed of a material having high thermal conductivity such as metal. For example, the module case **300** may be formed of aluminum or stainless steel. However, the material of the module case **300** is not limited thereto. The module case **300** may be referred to as a battery case, a housing, or a module housing.

[0069] The module case **300** may include an end plate **360**. The end plate **360** may surround at least a portion of the cell assembly **101**. In an embodiment, the end plate **360** may be connected to the accommodation portion **301** and/or an end portion of the module cover **302** in a length direction (for example, an X-axis direction). The end plate **360** may protect the cell assembly **101** from external impacts of the battery module **200**. The end plate **360** may cover a portion of a side surface of the cell assembly **101**. In an embodiment, the end plate **360** may include a protrusion portion **369** for coupling to an external structure of the battery module **200** (for example, a pack frame **410** of FIG. **10**). The end plate **360** may have a surface (for example, a third surface **360a**) oriented in the first direction (for example, a $+Z$ -direction). In an embodiment, the end plate **360** may be referred to as a front case and/or a rear case.

[0070] Components of the module case **300** may be coupled to each other using welding. For example, the module case **300** may include a first bonding portion **370** connecting the accommodation portion **301** and the module cover **302** to each other, a second bonding portion **380** connecting the accommodation portion **301** and the end plate **360** to each other, and/or a third bonding portion **390** connecting the module cover **302** and the end plate **360** to each other. In an embodiment, the bonding portion **370**, **380**, and **390** may be referred to as a welding line or a welding region. The first bonding portion **370**, the second bonding portion **380**, and the third bonding portion **390** may be formed to be substantially parallel or perpendicular to each other. A coupling structure of the module case **300** using the bonding portion **370**, **380**, and **390** will be described further below. In another embodiment, the module case **300** may include a fourth bonding portion **360** connecting the sidewall **320** and the end plate **360** of the accommodation portion **301** to each other. A direction in which the fourth bonding portion is arranged may be perpendicular to a direction in which the first bonding portion **370**, the second bonding portion **380**, and the third bonding portion **390** are arranged.

[0071] FIG. **5** is a side view of a battery module according to an embodiment. FIG. **6** is an enlarged view of region “A” of FIG. **5** according to an embodiment. FIG. **7A** is a perspective view of region “B” of FIG. **6** according to an embodiment. FIG. **7B** is a cross-sectional perspective view of region “B” of FIG. **6** according to an embodiment.

[0072] Referring to FIGS. **5**, **6**, **7A**, and/or **7B**, the battery module **200** may include a module case **300** including an accommodation portion **301** and a module cover **302**. At least a portion of the descriptions for the battery module **200**, the module case **300**, the accommodation portion **301**, and the module cover **302** of FIGS. **2**, **3**, and/or **4** may be applied to the battery module **200**, the module case **300**, the accommodation portion **301**, and the module cover **302** of FIGS. **5**, **6**, **7A**, and/or **7B**.

[0073] The accommodation portion **301** may be connected to the module cover **302**. For example, a sidewall **320** of the accommodation portion **301** may be bonded to a sidewall cover **340** of the module cover **302**. A module case **300** may include a first bonding portion **370** connecting the sidewall **320** and the sidewall cover **340** to each other. The upper plate **330** may have a first surface **330a** oriented in a first direction ($+Z$ -direction). The sidewall cover **340** may have a second surface

340a oriented in a second direction ($-Z$ -direction). The second surface **340a** may extend in a third direction (X -axis direction). The first bonding portion **370** may be welded to the second surface **340a** of the sidewall cover **340** and the sidewall **320** of the accommodation portion **301**. The first bonding portion **370** may be formed in the third direction (X -axis direction), parallel to the second surface **340a**. In an embodiment, the first bonding portion **370** may be referred to as a first welding line or a first welding bead.

[0074] Each of the sidewall **320** and the sidewall cover **340** may have a step structure. For example, the sidewall **320** may include a first protruding region **327** having an upper surface **320a** oriented in the first direction ($+Z$ -direction), and an internal region **326** positioned in the second direction ($-Z$ -direction) with respect to the first protruding region **327**. A portion of the first protruding region **327** may be in contact with the second surface **340a** of the sidewall cover **340**. For example, a portion of the first protruding region **327** may be bonded to the second surface **340a** of the sidewall cover **340** to form a first bonding portion **370**. The other portion of the first protruding region **327** may be spaced apart from a portion (for example, a shade forming surface **340b**) of the sidewall cover **340** by a guide groove **350**. The other portion of the first protruding region **327**, spaced apart from the shadow forming surface **340b**, may be referred to as a guide region **328**.

[0075] The module cover **340** may be formed to have a shape corresponding to a shape of the sidewall **320**. For example, the module cover **340** may include a second protruding region **347** opposing the internal region **326**. The second protruding region **347** may further protrude in the second direction ($-Z$) than the second surface **340a**. Each of the sidewall **320** and the sidewall cover **340** may have a protruding structure, thereby improving the ease of assembling the sidewall **320** and the sidewall cover **340**. A position of the first bonding portion **370** may be determined based on a contrast difference between the sidewall **320** of the accommodating portion **301** and the sidewall cover **340** of the module cover **302**. For example, an operator or a welding robot may sense the sidewall **320** and the sidewall cover **340**, and may determine, based on a contrast difference between the sidewall **320** and the sidewall cover **340**, a position of the first bonding portion **370**. The first bonding portion **370** may be positioned at a boundary between the sidewall **320** and the sidewall cover **340**.

[0076] In an embodiment, a material of the accommodation portion **301** may be the same as a material of the module cover **302**. For example, each of the sidewall **320** of the accommodation portion **301** and the sidewall cover **340** of the module cover **302** may include stainless steel or aluminum.

[0077] The module cover **302** may guide, to the operator, a position in which the first bonding portion **370** is to be formed. For example, the module cover **302** may include a guide groove **350**. The guide groove **350** may generate a contrast difference between the sidewall **320** and the sidewall cover **340**. For example, the guide groove **350** may be formed in the second surface **340a** of the sidewall cover **340**. For example, the guide groove **350** may be a groove formed in the second surface **340a** of the sidewall cover **340**. In an embodiment, the sidewall cover **340** may have a shade forming surface **340b** parallel to the second surface **340a**, the shade forming surface **340b** forming at least a portion of the guide groove **350**. The shade forming surface **340b** may be spaced apart from the first bonding portion **370**. The shade forming surface **340b** of the sidewall cover **340** may form a shade difference in a portion of the sidewall **320**. For example, the sidewall cover **340** in which the guide groove **350** is formed may generate shade in a portion of the sidewall **320**. In an embodiment, the sidewall **320** may include a guide region **328** formed by the sidewall cover **340**, and an exposed region **329** not covered by the sidewall cover **340**. At least a portion of the guide region **328** may be exposed to the outside of the module case **300** through the guide groove **350**. A brightness of the guide region **328** may be different from a brightness of the exposed region **329**. For example, a first brightness of the guide region **328** may be lower than a second brightness of the exposed region **329** by a shadow formed by the sidewall cover **340**. The operator may set,

based on a brightness difference between the guide region **328** and the exposed region **329**, a position of the first bonding portion **370**.

[0078] The guide groove **350** may be formed to have a shape for generating a contrast difference. For example, a width (for example, a first length **d1**) of the guide groove **350** may be formed to have a specified length (for example, 1 mm or less). The first length **d1** may be a length (for example, a second length **d2**) of the shade forming surface **340b** of the sidewall cover **340** in the third direction. A depth (for example, the second length **d2**) of the guide groove **350** may be formed to have a specified length (for example, 1 mm or less). The second length **d2** may be a length between the second surface **340a** and the shade forming surface **340b** of the sidewall cover **340**. A width (for example, a third length **d3**) of the guide groove **350** may be formed to have a specified length (for example, 0.5 mm or more). The third length **d3** may be a length of the shade forming surface **340b** of the sidewall cover **340** in a fourth direction (for example, a Y-axis direction). The guide groove **350** may be provided as a plurality of guide grooves **350**. For example, the guide groove **350** may include a first guide groove **351**, a second guide groove **352**, and a third guide groove **353**, spaced apart from each other. The plurality of guide grooves **351**, **352**, and **353** may be arranged in the third direction (for example, the X-axis direction). The plurality of guide grooves **350** may be provided, thereby increasing positional accuracy of the first bonding portion **370**. The number of guide grooves **350** may be selectively designed.

[0079] The first bonding portion **370** may be provided as a plurality of first bonding portions **370**. For example, the first bonding portion **370** may include a plurality of first bonding portions **371** and **372**, spaced apart from each other with respect to the guide groove **350** and/or the guide region **328**. The guide region **328** may be positioned between the plurality of first bonding portions **371** and **372**. The number of first bonding portions **370** may be selectively designed.

[0080] The battery module **200** may receive pressure from a pressurizing device positioned in the first direction (+Z-direction) with respect to the module cover **302**. For example, the battery module **200** may receive pressure from a pressurizing device positioned in the first direction (+Z-direction) with respect to the module cover **302** in a state of being supported by a support (for example, an assembly jig) positioned in the second direction (−Z-direction) with respect to the accommodating portion **301**. A distance between the module cover **302** and the sidewall **320** may be reduced by the pressurizing device. For example, the first bonding portion **370** may be formed on the sidewall cover **340** and the sidewall **320** in a state in which the distance between the sidewall cover **340** and the sidewall **320** of the module cover **302** is reduced. The distance between the module cover **302** and the sidewall **320**, being reduced, may improve durability of the battery module **200**.

[0081] FIG. **8** is a cross-sectional view taken along line A-A' of FIG. **2** according to an embodiment.

[0082] Referring to FIG. **8**, the battery module **200** may include a cell assembly **101** and a module case **300** accommodating the cell assembly **101**. The module case **300** may include a main plate **310** and an end plate **360**.

[0083] At least a portion of the descriptions of the cell assembly **101**, the module case **300**, the main plate **310**, and the end plate **360** of FIGS. **2**, **3**, and/or **4** may be applied to the cell assembly **101**, the module case **300**, the main plate **310**, and the end plate **360** of FIG. **8**.

[0084] The cell assembly **101** may include a plurality of battery cells (for example, the battery cell **100** of FIG. **1**). The cell assembly **101** may include an insulating member **160** for preventing contact between the battery cell **100** and the end plate **360**.

[0085] The module case **300** may include a second bonding portion **380** connecting the main plate **310** and the end plate **360** of the accommodation portion **301** to each other. The end plate **360** may be fixed to the main plate **310** by the second bonding portion **380**.

[0086] The module case **300** may have a shape for preventing damage by the second bonding portion **380**. The main plate **310** and the end plate **360** may have a step structure. For example, the

main plate **310** may include a first central portion **311** formed to have a first thickness t_1 , and a first end portion **312** extending from the first central portion, the first end portion **312** formed to have a second thickness t_2 , less than the first thickness t_1 . The end plate **360** may include a second central portion **361** formed to have a third thickness t_3 , and a second end portion **362** extending from the second central portion, the second end portion **362** formed to have a fourth thickness t_4 , less than the third thickness t_3 . The second bonding portion **380** may connect the first end portion **312** of the main plate **310** and the second end portion **362** of the end plate **360** to each other.

[0087] The second bonding portion **380** may be formed in a welding direction WD, inclined with respect to the first end portion **312** of the main plate **310** and the second end portion **362** of the end plate **360**. For example, the first end portion **312** may have a first internal surface **313** having at least a portion opposing the second central portion **361**, and a first bonding surface **314** perpendicular to the first internal surface **313**. The second end portion **362** may have a second internal surface **363** opposing the first central portion **311**, and a second bonding surface **364**. The second bonding surface **364** may be perpendicular to the second internal surface **363** and the first bonding surface **314**. The second bonding portion **380** may connect the first bonding surface **314** and the second bonding surface **364** to each other. The second bonding portion **380** may be formed in a welding direction WD, inclined to the first bonding surface **314** and the second bonding surface **364**. For example, the welding direction WD may be inclined by about 45 degrees with respect to the first bonding surface **314** and the second bonding surface **364**. The second bonding portion **380** may be formed by melting and hardening a contact surface between the first bonding surface **314** and the second bonding surface **364** in the welding direction WD.

[0088] The second bonding portion **380** may be disposed on the first bonding surface **314** of the main plate **310** and the second bonding surface **364** of the end plate **360**, thereby preventing damage to the battery cell **100** and other portions (for example, the central portion **311** and **361**) of the module case **300** due to heat and/or laser generated during welding. For example, each of the main plate **310** and the end plate **360** may have a step structure, and the second bonding portion **380** may be inclinedly transferred with respect to the bonding surfaces **314** and **364**, thereby preventing heat transfer to an unintended portion. In an embodiment, the second bonding portion **380** may be referred to as a second welding line or a second welding bead.

[0089] The battery module **200** may receive pressure from a pressurizing device positioned in a first direction (+Z-direction) with respect to the end plate **360**. For example, the battery module **200** may receive pressure from a pressurizing device positioned in the first direction (+Z-direction) with respect to the end plate **360** in a state of being supported by a support (for example, an assembly jig) positioned in the second direction (−Z-direction) with respect to the main plate **310**. A distance between the main plate **310** and the end plate **360** may be reduced by the pressurizing device. For example, the second bonding portion **380** may be formed on the main plate **310** and the end plate **360** in a state in which the distance between the main plate **310** and the end plate **360** of the accommodation portion **301** is reduced. The distance between the main plate **310** and the end plate **360**, being reduced, may improve durability of the battery module **200**.

[0090] In the present disclosure, it is illustrated that the main plate **310** and the end plate **360** have a step structure, and the second bonding portion **380** connects the main plate **310** and the end plate **360** having the step structure to each other, but a step structure of the module case **300** is not limited to the main plate **310** and the end plate **360**. For example, the step structures of the main plate **310** and the end plate **360** may be applied to other components of the module case **300**. In another embodiment (not illustrated), a third bonding portion (for example, the third bonding portion **390** of FIG. 2) may be connected to an upper plate (for example, the upper plate **330** of FIG. 2) and the end plate **360**, having a step structure, to each other. In another embodiment (not illustrated), each of a sidewall (for example, the sidewall **320** of FIG. 2) and the end plate **360** may have the step structure, and the fourth bonding portion (not illustrated) may connect the sidewall **320** and the end plate **360**, having a step structure, to each other.

[0091] FIG. 9 is a top view of a battery module according to an embodiment.

[0092] Referring to FIG. 9, a battery module **200** may include an upper plate **330** and an end plate **360**. At least a portion of the descriptions of the battery module **200**, the upper plate **330**, and the end plate **360** of FIGS. 2, 3, and/or 4 may be applied to the battery module **200**, the upper plate **330**, and the end plate **360** of FIG. 9.

[0093] The module case **300** may include a third bonding portion **390** connecting the upper plate **330** and the end plate **360** to each other. The upper plate **330** may have a substantially rectangular plate shape. For example, the upper plate **330** may include a first end region **331** connected to a sidewall cover (for example, the sidewall cover **340** of FIG. 4), and a second end region **332** perpendicular to the first end region **331**. The third bonding portion **390** may connect the second end region **332** of the upper plate **330** and the end plate **360** to each other. In an embodiment, the third bonding portion **390** may be referred to as a third welding line or a third welding bead.

[0094] The third bonding portion **390** may connect the upper plate **330** and the end plate **360** to each other at a plurality of points. For example, the upper plate **330** may include a plurality of second end regions **332a** and **332b**. The plurality of second end regions **332a** and **332b** may include a second-first end region **332a** and a second-second end region **332b**, oriented in opposite directions. The third bonding portion **390** may include a third-first bonding portion **391** connecting the second-first end region **332a** and the end plate **360** to each other, and a third-second bonding portion **392** connecting the second-second end region **332b** and the end plate **360** to each other.

[0095] The third bonding portion **390** may be positioned upwardly (for example, in a first direction (+Z-direction)) with respect to the upper plate **330** and the end plate **360**. For example, the end plate **360** may have a third surface **360a** oriented in the first direction (+Z-direction). The bonding portion **390** may connect the third surface **360a** of the end plate and the first surface **330a** of the upper plate **330** to each other.

[0096] The battery module **200** may receive pressure from a pressurizing device positioned in a third direction (X-axis direction) with respect to the end plate **360**. For example, the battery module **200** may receive pressure from a pressurizing device and/or a supporting device in contact with the end plates **360**. A distance between the end plate **360** and the upper plate **330** of the module cover **302** and a distance between the end plate **360** and the accommodation portion **301** may be reduced by the pressurizing device (or the supporting device). For example, the third bonding portion **390** may be formed on the upper plate **330** and the end plate **360** in a state in which the distance between the upper plate **330** and the end plate **360** is reduced. The distance between the upper plate **330** and the end plate **360**, being reduced, may improve durability of the battery module **200**.

[0097] The third bonding portion **390** may be positioned in the first direction (+Z-direction) with respect to the upper plate **330** and the end plate **360**, such that a distance between components (for example, the module case **300** of FIG. 4) of the battery module **200** may be reduced. For example, the third bonding portion **390** may be formed independently of a first bonding portion (for example, the first bonding portion **370** of FIG. 6) or a second bonding portion (for example, the second bonding portion **380** of FIG. 8).

[0098] The third bonding portion **390** may be formed on components (for example, the upper plate **330** and the end plate **360**) pressed in the third direction (X-axis direction). The first bonding portion **370** may be formed on components (for example, the module cover **302** and the sidewall **320**) pressed in a second direction (for example, the -Z-direction of FIG. 4), perpendicular to the third direction (X-axis direction). The second bonding portion **380** may be formed on components (for example, the main plate **310** and the end plate **360**) pressed in the second direction (for example, the -Z-direction of FIG. 4), perpendicular to the third direction (X-axis direction).

[0099] The third bonding portion **390** may be positioned in the first direction (+Z-direction) with respect to the upper plate **330** and the end plate **360**, such that a distance between the upper plate **330** and the end plate **360** may be reduced by a pressurizing device in a state in which an distance between the module cover **302** and the sidewall **320** and a distance between the main plate **310** and

the end plate **360** are reduced. The distance between the components of the battery module **200** (for example, the module case **300** of FIG. **4**) may be reduced, thereby improving durability of the battery module **200**.

[0100] FIG. **10** is an exploded perspective view of a battery pack according to an embodiment.

[0101] Referring to FIG. **10**, a battery pack **400** may include a plurality of battery modules **200** and a pack frame **410** accommodating the plurality of battery modules **200**. The above descriptions (for example, FIGS. **2** and **3**) of the battery module **200** may be applied to the battery module **200** of FIG. **10**. For example, the battery module **200** of FIG. **10** may include a module case (for example, the module case **300** of FIG. **4**).

[0102] The pack frame **410** may accommodate a component (for example, the battery module **200**) of the battery pack **400**. The pack frame **410** may include a bottom member **411** supporting the battery module **200**, a pack cover **412** covering the battery module **200**, and a pack sidewall **413** surrounding the bottom member **411** and at least a portion of the pack cover **412**. The bottom member **411** may support a case (for example, the module case **300** of FIG. **2**) of the battery module **200**.

[0103] The pack frame **410** may include a partition wall **420** crossing at least some battery modules **200**, among the plurality of battery modules **200**. For example, an accommodation space of the pack frame **410** may be divided into a plurality of spaces by the partition wall **420**. The partition wall **420** may be installed to cross the accommodation space so as to increase the rigidity of the pack frame **410**. In an embodiment, the partition wall **420** may include a first partition wall **420a** crossing a plurality of battery cells **100**, and a plurality of second partition walls **420b** substantially perpendicular to the first partition wall **420a**.

[0104] In an embodiment, the battery pack **400** may include a duct member **430**. The duct member **430** may include an exhaust space for providing a path of gas and/or flame discharged from the battery module **200**. The duct member **430** may be disposed in the pack frame **410**. The duct member **430** may surround at least a portion of the battery module **200**. For example, gas and/or flame, generated in a battery cell (for example, the battery cell **100** of FIG. **1**) of the battery module **200**, may pass through the exhaust space of the duct member **430** and be transferred to the outside of the battery pack **400**. In the present disclosure, the duct member **430** may be referred to as an exhaust duct or an exhaust member.

[0105] The battery pack **400** may include a battery control unit **490**, controlling the battery module **200**. The battery control unit **490** may be disposed in the pack frame **410**. The battery control unit **490** may include a battery management system (BMS). A configuration of the battery control unit **490** is well known in various forms, such that detailed descriptions thereof are omitted. In an embodiment, the battery control unit **490** may be referred to as a processor.

[0106] A structure of the battery pack **400** of FIG. **10** is an example. For example, the number of battery modules **200** included in the battery pack **400**, a structure of the pack frame **410**, and/or a structure of the duct member **430** may be selectively designed.

[0107] The features described above are merely examples of the application of the principles of the present disclosure, and other components may be included without departing from the scope of the present disclosure.

[0108] Only specific examples of implementations of certain embodiments are described.

Variations, improvements and enhancements of the disclosed embodiments and other embodiments may be made based on the disclosure of this patent document. For example, the present disclosure may be implemented by deleting some components from the above-described embodiments, and respective embodiments may be implemented in combination with each other.

Claims

- 1.** A battery module comprising: a cell assembly including a plurality of battery cells; and a module case accommodating the cell assembly, wherein the module case includes: an accommodation portion including a main plate supporting the cell assembly and a sidewall extending from the main plate; a module cover including an upper plate covering the cell assembly, the upper plate having a first surface oriented in a first direction, a sidewall cover extending from the upper plate, the sidewall cover covering a portion of the sidewall, the sidewall cover having a second surface oriented in a second direction opposite to the first direction, and a guide groove formed in the second surface; and a first bonding portion connecting the second surface of the sidewall cover and the sidewall to each other.
- 2.** The battery module of claim 1, wherein the sidewall cover has a shade forming surface parallel to the second surface, the shade forming surface forming at least a portion of the guide groove, and the shade forming surface is spaced apart from the first bonding portion.
- 3.** The battery module of claim 2, wherein the sidewall includes: a guide region formed by the sidewall cover, the guide region having at least a portion exposed through the guide groove; and an exposed region spaced apart from the guide region, the exposed region not covered by the module cover, wherein the guide region has a first brightness, and the exposed region has a second brightness different from the first brightness.
- 4.** The battery module of claim 3, wherein the first bonding portion includes a plurality of first bonding portions, and the guide region is disposed between the plurality of first bonding portions.
- 5.** The battery module of claim 1, wherein a material of the module cover and a material of the accommodation portion are the same.
- 6.** The battery module of claim 1, wherein the module case includes an end plate covering at least a portion of the cell assembly, the end plate perpendicular to the sidewall, the main plate includes a first central portion having a first thickness, and a first end portion extending from the first central portion, the first end portion has a second thickness less than the first thickness, the end plate includes a second central portion has a third thickness, and a second end portion extending from the second central portion, the second end portion has a fourth thickness less than the third thickness, and the module case includes a second bonding portion connecting the first end portion and the second end portion to each other.
- 7.** The battery module of claim 6, wherein at least a portion of the first end portion has a first internal surface opposing the second central portion, and a first bonding surface perpendicular to the first internal surface, the second end portion has a second internal surface opposing the first central portion, and a second bonding surface perpendicular to the second internal surface and the first bonding surface, and the second bonding portion connects the first bonding surface and the second bonding surface to each other.
- 8.** The battery module of claim 7, wherein the second bonding portion is formed in a welding direction, inclined with respect to the first bonding surface and the second bonding surface.
- 9.** The battery module of claim 1, wherein the upper plate includes a first end region connected to the sidewall cover, and a second end region perpendicular to the first end region, and the module case includes an end plate covering at least a portion of the cell assembly, the end plate is perpendicular to the sidewall, and a third bonding portion connecting the second end region of the upper plate and the end plate to each other.
- 10.** The battery module of claim 9, wherein the end plate has a third surface oriented in the first direction, and the third bonding portion connects the first surface of the upper plate and the third surface of the end plate to each other.
- 11.** A battery module comprising: a cell assembly including a plurality of battery cells; and a module case accommodating the cell assembly, wherein the module case includes: an accommodation portion including a main plate supporting the cell assembly, and a sidewall extending from the main plate; a module cover including an upper plate covering the cell assembly,

the upper plate having a first surface oriented in a first direction, and a sidewall cover extending from the upper plate, the sidewall cover covering a portion of the sidewall, the sidewall having a second surface oriented in a second direction opposite to the first direction; an end plate surrounding at least a portion of the cell assembly, the end plate having a third surface oriented in the first direction; a first bonding portion connecting the second surface of the sidewall cover and the sidewall to each other; a second bonding portion connecting the main plate and the end plate to each other; and a third bonding portion connecting the first surface of the upper plate and the third surface of the end plate to each other.

12. The battery module of claim 11, wherein the module cover includes a guide groove formed in the second surface, the sidewall cover has a shade forming surface parallel to the second surface, the shade forming surface forming at least a portion of the guide groove, and the shade forming surface is spaced apart from the first bonding portion.

13. The battery module of claim 11, wherein the end plate is perpendicular to the sidewall, the main plate includes a first central portion having a first thickness, and a first end portion extending from the first central portion, the first end portion has a second thickness less than the first thickness, the end plate includes a second central portion having a third thickness, and a second end portion extending from the second central portion, the second end portion has a fourth thickness less than the third thickness, and the second bonding portion connects the first end portion and the second end portion to each other.

14. The battery module of claim 13, wherein at least a portion of the first end portion has a first internal surface opposing the second central portion, and a first bonding surface perpendicular to the first internal surface, the second end portion has a second internal surface opposing the first central portion, and a second bonding surface perpendicular to the second internal surface and the first bonding surface, the second bonding portion connects the first bonding surface and the second bonding surface to each other.

15. A battery pack comprising: a plurality of battery modules; and a pack frame accommodating the plurality of battery modules, wherein each of the plurality of battery modules includes: a cell assembly including a plurality of battery cells; and a module case accommodating the cell assembly, and the module case includes: an accommodation portion including a main plate supporting the cell assembly, and a sidewall extending from the main plate; a module cover including an upper plate covering the cell assembly, the upper plate having a first surface oriented in a first direction, a sidewall cover extending from the upper plate, the sidewall cover covering a portion of the sidewall, the sidewall cover having a second surface oriented in a second direction opposite to the first direction, and a guide groove formed in the second surface; and a first bonding portion connecting the second surface of the sidewall cover and the sidewall to each other.
